

BorderEye - SIH Brief: Tech Stack, AI, Blockchain & App Workflow

Border Intelligence & Video Analytics Engine | Smart India Hackathon 2026

PART 1 - TECH STACK (SIH PPT)

1.1 Layer-wise stack

Dashboard UI | Streamlit 1.58

Full command-centre UI: live feed, alert log, sidebar, analytics, tripwire canvas. Python-only.

Drawing input | streamlit canvas lib

Pen-type tripwire designer - click-to-draw line input on canvas.

Detection AI | YOLOv8 (Ultralytics 8.4.150)

Detects person + 6 vehicle classes in every frame.

Tracking AI | ByteTrack (YOLO built-in)

Persistent ID per object (PERSON #3), follows frame-to-frame.

Custom weights | detection.pt (IDD fine-tuned)

Fine-tuned on Indian roads: person, bicycle, car, motorcycle, bus, truck, autorickshaw.

Plate AI | plate.pt + EasyOCR 1.7.2

Number-plate region proposals + OCR text reading (ANPR).

Face detection | OpenCV Haar cascade 4.14

Blue boxes on faces (classical ML, basic accuracy).

Image pipeline | OpenCV 4.14 + NumPy, Torch 2.13 CPU

Resize, night-enhance (CLAHE), video I/O, inference - all on CPU, no GPU needed.

API backend | FastAPI (src/backend/api.py)

/api/tracks and detection endpoints - C2-system integration contract.

AI summarizer | Groq API (gpt-oss-20b)

Alert metadata to 2-line English summary + Hindi action line.

Ledger/crypto | SHA-256 hash chain (stdlib)

Every alert linked to previous - tamper-evident edge ledger + anchor log.

Reports | fpdf2

One-click English PDF incident report from sidebar, works offline.

Config/theme | TOML theme + secrets

Dark tactical theme; API key in local-only secrets.toml.

Language | Python 3.14

Single-language stack - from demo to deployment.

1.2 AI - where and how (judges' favourite slide)

- Detection - YOLOv8 IDD-finetuned model, confidence 0.25 (recall-first for borders), 7 classes.
- Tracking - model.track (persist=True), ByteTrack IDs; loitering (15s dwell), tripwire-crossing (segment-intersection on track path), fast-movement (px/frame jump) - all built on track IDs.
- ANPR - plate-shape contour filter -> EasyOCR -> text + confidence.
- Face detection - Haar boxes (honest label: basic).
- LLM layer - Groq summarizer (demo/HQ mode only - metadata leaves the device; put this caveat on the slide).

1.3 Blockchain - where and how

- Edge Ledger: every alert is a block (prev_hash -> hash, SHA-256). Edit any old alert and the chain breaks - the Verify

button demos exactly this, live.

- Anchoring: 'Anchor Now' creates a self-chained checkpoint of the day's head-hash (anchors.jsonl); add a timestamp URL (Polygon Amoy relay) for public proof + HTTP receipt. One-liner for PPT: 'Private edge ledger for real-time, public anchor for audit.'

- Honest scope: no full on-chain-per-alert - slow, costly, useless in offline border areas. Say this.

PART 2 - FULL APP WORKFLOW (IN DETAIL)

0. Login gate. The entire app sits behind a demo login (any email + exactly 6-digit password). Nothing - no sidebar, no feed - renders until authed is true. Logout wipes the session.

1. Header. BorderEye logo + title, + Add Camera/Footage popover (RTSP save with name/location, or video upload), link banner ([OFFLINE] edge-mode warning), SOS banner when active, on-device analytics notice.

2. Live Detection Feed (4 sources). Demo Mode (synthetic feed), Upload Video (up to 4 videos in tabs, Play with progress bar + stride control, or Single-Frame inspect with OCR overlay), Webcam (snapshot/live), RTSP Stream (connect/disconnect, live tracking). Every frame passes through: night-enhance (if night) -> YOLO detect + ByteTrack -> fence check -> tripwire check -> face boxes -> HUD overlay (counts, frame no.).

3. Alert Log. Every incident card shows: severity icon, event type, threat score (deterministic 0-100 with expandable 'why'), explanation, timestamp + event ID (mono), and sync status ([OK] synced / [QUEUED] queued).

4. Risk scoring. Severity base + detection confidence + fence breach + night hours = a documented formula, not a black box.

5. Dashboard metrics + Latest Alert Object (full JSON) + Low-Bandwidth Payload preview (exact bytes that travel - video never does) + AI Summary (Groq, after sidebar action) + Hash-chain Verify.

6. Behavioral Analytics. Frames analyzed, persons/vehicles/faces seen, class bar chart, hourly activity chart, reset.

7. Sidebar. Model status, Night Mode toggle, Face Detection toggle, simulate buttons (intrusion/ANPR/signal-loss/restore/clear), Link toggle + outbox counts + Sync Now, SOS + HQ webhook + Stand Down, Footage library (4), RTSP library (4), Edge Ledger + Anchor + Verify Anchors, PDF report + download, AI Summarizer (key + N + Summarize), camera health list, user + logout.

8. Tripwire Designer. Click points on a 640x480 canvas (1:1 with detection frame) to sketch any free-form line -> Save as tripwire (magenta TW lines render on live feed) -> any tracked object crossing fires a tripwire_crossing alert with track ID. Undo/clear/delete-all included; coordinate fallback if canvas unavailable.

9. Offline flow (the SIH story). Link OFF -> AI keeps running, alerts queue in data/outbox.jsonl, video stays on device -> Link ON -> auto-sync with count, every card flips to [OK]. Only ~100-200 bytes of metadata ever cross the link.